SafeMAP: Analysis-Guided Planning and Per-Function Validation for Safe C-to-Rust Translation

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Abstract

Automatic C-to-Rust translators either keep C's pointer conventions in unsafe Rust or rely on a large language model (LLM) whose output is hard to check. SafeMAP is a conservative alternative. For each C function it first decides, by static analysis, whether a safe Rust interface exists; if not, it refuses the function and records why. For the remaining functions it emits Rust from exact rules or from a plan-guided LLM request with compiler-feedback repair, and it accepts a function only if the code contains no `unsafe`, exposes no raw pointers, and matches the original C function on up to 1,000 generated inputs, with the C side run under AddressSanitizer and UndefinedBehaviorSanitizer. We froze the implementation and evaluated it on ten held-out C libraries (9,456 lines, 356 translation units) selected by a sealed protocol. Across three runs, SafeMAP accepted 12–16 units and a one-shot LLM given the same planned interfaces 9–15; rules alone accepted 1 and raw C2Rust none. Most units were refused before translation: 159 need manual refactoring and 116 use C types with no safe, testable Rust interface. Mutation analysis shows that the per-function check rejects 89% of compiling single-operator mutants of accepted functions. Ablations on development libraries attribute the accepted translations mainly to safe-signature planning and idiom plans, not to prompt guidance or repair; with the planned interfaces, a single LLM request performs within the run-to-run range of the full loop. The results show that per-function, oracle-checked acceptance is feasible on production C but that coverage is low; the recorded refusals identify what currently blocks automatic safe migration.

Keywords: C-to-Rust migration, safe Rust, static analysis, large language models, differential testing, memory safety

1 Introduction

C remains common in systems software. Its memory model leaves programmers responsible for preventing buffer overflows, use-after-free errors, null dereferences, and data races. Safe Rust checks ownership and borrowing at compile time while retaining low-level control, so migrating C code to safe Rust is attractive. Translation is difficult because many C idioms have no direct safe-Rust equivalent: raw pointers, output parameters, manual allocation, nullable values, and integer error codes all require changes in representation and interface.

C2Rust preserves C behavior by generating Rust that uses raw pointers and unsafe operations Immunant (n.d.). Later work reduces this unsafety by static analysis, ownership inference, refactoring, and LLM rewriting Emre et al. (2021); Hong (2023); Zhang et al. (2023); Ling et al. (2022); Nitin et al. (2026); Mohapatra et al. (2026). A second line asks LLMs to translate directly to safe Rust and checks the result with tests, fuzzing, or verification Hong and Ryu (2025b); Zhou et al. (2025a); Eniser et al. (2025); Shetty et al. (2024); Zhou et al. (2025b); Yang et al. (2025); Sim et al. (2026). These systems generally aim to translate everything and then measure how much of the result is safe or correct.

SafeMAP takes the opposite position. It translates a function only when static analysis finds a safe Rust interface for it, and it accepts a translation only when that function, on its own, passes a behavioral check against the original C. Everything else is refused with a recorded reason. The question is therefore not how much unsafe code remains, but how many functions of real C code can be migrated to safe Rust with per-function evidence, and what prevents the rest.

An earlier, frozen evaluation of SafeMAP on three C libraries accepted no units: defects in interface planning, rule matching, and harness construction each forced the outcome to zero (Section 4.1). This paper reports the corrected system, evaluated under a new sealed protocol on ten libraries that were not used during development. We ask:

- **RQ1:** How many translation units of production C libraries does SafeMAP accept as safe, behaviorally validated Rust?
- **RQ2:** Why are the remaining units not accepted?
- **RQ3:** How does SafeMAP compare with a one-shot LLM given the same interfaces, with its own rules alone, and with raw C2Rust, under one acceptance policy?
- **RQ4:** Which planning components contribute to acceptance?
- **RQ5:** How reliable is the acceptance decision, in terms of the harness's ability to reject wrong code and the variation across LLM runs?

This paper contributes:

- a translation pipeline that decides, per function and before any translation, whether a safe and testable Rust interface exists, and records a reason when it does not;
- a per-function acceptance procedure that compares each Rust function with the original C function under sanitizers, excluding inputs on which the C code has undefined behavior;
- a sealed, frozen evaluation on ten held-out libraries with a one-shot LLM baseline that receives the same interfaces, repeated LLM runs, mutation analysis of the acceptance check, and component ablations; and
- a breakdown of 356 held-out units by the reason they could not be migrated, together with a public artifact for every decision.

2 Background and related work

2.1 C2Rust and safer-Rust refactoring

C2Rust translates C through a mostly syntactic process and is often used as a baseline or a migration starting point Immunant (n.d.). It preserves many C constructs with raw pointers and unsafe operations, so much of its output falls outside Rust's safe subset. Emre et al. found that raw pointers cause much of this unsafety Emre et al. (2021). Hong used static analysis to replace selected unsafe constructs in C2Rust output, including lock APIs and output parameters Hong (2023). C2SaferRust combines C2Rust, static analysis, and LLM rewriting to reduce unsafe code in translated real-world programs Nitin et al. (2026). Crown infers ownership models for C pointers and converts many of them to safe Rust types Zhang et al. (2023). CRustS and Cp2SRust use different source-to-source transformations and transpiler designs Ling et al. (2022); Mohapatra et al. (2026). Industry studies and surveys report that automatic translation still needs substantial manual work and testing Hong and Ryu (2025a); Viirola et al. (2025).

2.2 LLM-based C-to-Rust translation

LLMs can produce idiomatic Rust, although correctness and type migration remain open problems. Hong and Ryu studied an LLM translator supplied with candidate signatures, caller and callee context, and compiler feedback Hong and Ryu (2025b). C2RustTV checks LLM translations with generated tests and repair loops Zhou et al. (2025a). Eniser et al. evaluate LLM translation through cross-language differential fuzzing and find lower success on more complex real-world samples Eniser et al. (2025). RustFlow studies prompting and iterative repair Zhang et al. (2026). Valenzuela et al. examine compiler errors in direct C-to-safe-Rust translation Valenzuela et al. (2025). RustMap combines program analysis and LLM support at project scale Cai et al. (2025).

Several recent systems validate LLM translations against the C program. Syzygy translates units in dependency order and tests each one against its C counterpart with LLM-generated inputs Shetty et al. (2024). SACTOR first produces interface-preserving Rust that may use `unsafe`, then refines it to idiomatic safe Rust, and checks both stages with end-to-end tests linked to the C code through the foreign-function

Table 1 Validation and scope of recent C-to-Rust systems, as described by their authors.

System	Output	Behavioral oracle	Code it cannot translate safely
C2Rust Immunant (n.d.)	unsafe Rust	none	translated unsafely
Syzygy Shetty et al. (2024)	safe Rust	C, LLM-generated inputs	repaired by LLM
SACTOR Zhou et al. (2025b)	safe Rust	end-to-end tests via FFI	unsafe first stage
VERT Yang et al. (2025)	safe Rust	Wasm-derived Rust oracle	regenerated
ENCRUST Sim et al. (2026)	safe Rust	project test vectors	shims, then agent
SafeMAP	safe Rust	C under ASan/UBSan, per function	refused with reason

interface Zhou et al. (2025b). VERT checks a readable LLM translation against a WebAssembly-derived oracle with bounded model checking or property-based testing Yang et al. (2025). ENCRUST wraps LLM translations in ABI-preserving shims and removes the remaining unsafe code with an agent Sim et al. (2026), and Reboot translates whole interpreters to safe Rust through feature milestones with occasional user intervention Wang et al. (2026). CRUST-Bench supplies 100 C repositories with hand-written safe-Rust interfaces and tests Khatry et al. (2025); those interfaces play the role that SafeMAP's planned signatures play here.

Table 1 places SafeMAP among these systems. SafeMAP differs in three ways. It decides per function, before translation, whether a safe interface exists, and records a reason when it does not instead of emitting `unsafe` code. Its oracle is the original C function itself, run under ASan and UBSan, and inputs on which C has undefined behavior are excluded rather than used as expected outputs. And acceptance is per function, so one failing unit does not hide the others. The cost of this conservatism is coverage, which Section 5 reports.

2.3 Undefined behavior and the C oracle

Differential testing needs a trustworthy oracle. A C execution with undefined behavior does not define an expected output. ASan and UBSan detect many such executions Serebryany et al. (2012); LLVM Project (n.d.); SafeMAP runs every C oracle execution under both and discards inputs on which either reports an error. Miri can detect undefined behavior in Rust Jung et al. (2026), but generated code that contains no `unsafe` cannot exhibit it, so SafeMAP does not run Miri.

3 SafeMAP design

3.1 Acceptance policy

A translation unit is a C function, or a group of mutually recursive functions. SafeMAP accepts a unit if and only if all of the following hold:

1. it was admitted to the crate: it has the planned signature, contains no `unsafe`, raw pointers, `extern` blocks, or placeholders, and compiles together with the units already admitted;
2. the final crate compiles under `#![forbid(unsafe_code)]`, and its tests and Clippy pass;
3. its public interface exposes no raw pointer types; and
4. its per-unit differential test passes (Section 3.4).

A skipped or inapplicable check never counts as a pass. A unit whose callee was not admitted is recorded as `dependency_missing`.

3.2 Pipeline

Figure 1 shows the pipeline. For each C project, SafeMAP:

1. recovers compile information and analyzes every function with libclang: signatures, pointer roles (read, write, index, length pairing, nullability), allocation, calls, and unsupported constructs;
2. groups mutually dependent functions into units and orders them by dependency;
3. screens each unit (`candidate_safe`, `manual_refactor_required`, `unsafe_required`, `unknown`); mutable global state makes a unit `manual_refactor_required`;
4. plans a safe Rust signature for each candidate. The planner reports every C type it cannot map to a safe, testable Rust type: opaque `void *` or `FILE *`, pointer-to-pointer, returned aliasing pointers, and structs whose pointer fields have unresolved ownership. A unit with any such issue is refused at planning; no signature is fabricated. A struct pointer may be planned as a Rust struct of the struct's scalar fields when every use of the pointer in the unit is a scalar-field access, a null test, or an argument to a function that satisfies the same condition;
5. synthesizes each planned unit in dependency order, by an exact deterministic rule when one applies and otherwise by a plan-guided LLM request followed by up to three compiler-feedback repairs; and
6. validates the crate and each unit against the acceptance policy.

3.3 Deterministic rules and plan-guided synthesis

Deterministic rules cover scalar expressions, boolean and bit-reversal idioms, pointer-length and fixed-size arrays, scalar structs and field access, dependency-ordered internal calls, output parameters, return-code plus output-parameter to `Result`, nullable pointers, reductions, simple C string lengths, and simple allocation-to-`Vec` and `Box` patterns. A rule that renders a single expression applies only when the whole function body is that statement and the expression references only parameters and called functions.

When no rule applies, SafeMAP sends the LLM the C function, the macros, enums, and constant tables it references, the Rust definitions of the planned structs, the signatures of already-accepted callees, the analysis facts, and the migration plan, and asks for exactly the planned signature. The response is admitted only if it passes the

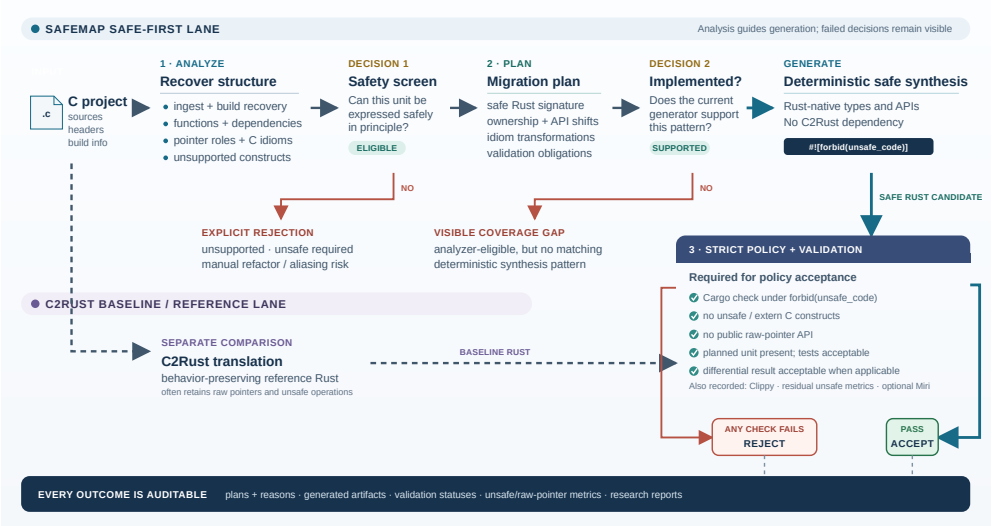


Fig. 1 SafeMAP conversion and acceptance pipeline. Static analysis records eligibility separately from synthesis support. Where no deterministic rule matches a planned unit, the generation step issues a plan-guided LLM request with compiler-feedback repair (not drawn). C2Rust runs in a separate reference lane. The same policy applies to every output, and reports retain refused units, failed checks, and their reasons

policy check and compiles with the units already admitted; otherwise the compiler diagnostics are returned for up to three repairs.

Consider the LLVM development helper below. libclang finds two indexed pointers paired with `n` and no writes, so the plan maps each pointer-length pair to a shared slice, and the `slice_dot_product` rule applies:

```
double loop(float *x, float *y, long n) {
  double acc = 0.0;
  for (long i=0; i<n; ++i) acc += (double)x[i] * (double)y[i];
  return acc;
}
```

```
pub fn r#loop(x: &[f32], y: &[f32]) -> f64 {
  x.iter().zip(y.iter()).map(|(&a,&b)| (a as f64)*(b as f64))
   .fold(0.0_f64, |s,v| s + v)
}
```

The initial value `+0.0` is necessary: differential testing showed that `sum()` can disagree with C on the sign of zero.

3.4 Per-unit differential harness

For each admitted unit, SafeMAP generates inputs from a fixed seed, including integer literals mined from the C body and their neighbors, and embeds the same literal tables in a C harness and a Rust harness. Slice and array arguments are exact-size heap buffers, so an out-of-bounds read in C is caught by ASan rather than masked. The C harness includes the original translation unit and runs under ASan and UBSan with

`-fno-sanitize-recover=all`; the Rust harness is built with overflow checks. Return values and every value written through a mutable parameter are compared exactly (floating point by bit pattern, strings byte by byte). An input on which the C oracle stops with a sanitizer report has no defined behavior and is excluded and counted. A defined C `abort` or `exit` must be matched by a Rust panic. A unit needs at least 50% defined inputs; otherwise it is recorded as `c_oracle_undefined_behavior`.

3.5 Baselines

Rules only runs the same pipeline without the LLM. *Raw C2Rust* translates each project in a pinned container (C2Rust 0.22.1, LLVM 14) and is measured under the same policy. The *one-shot LLM* baseline is given the same task as SafeMAP in a single request: the project's C sources and headers, the planned public signatures (C names kept), and the Rust struct definitions those signatures use, with private helpers allowed. It receives none of SafeMAP's analysis facts, plans, rules, or repair. Its output is scored with the same per-unit harness after making top-level functions public. A baseline that is not told the interfaces cannot be tested through them, so giving them isolates what SafeMAP adds beyond interface planning; the contribution of interface planning itself is measured by ablation.

4 Evaluation methodology

4.1 Earlier evaluation

A first frozen evaluation (protocol v1) ran SafeMAP on inih, cJSON, and libcsv and accepted 0 of 142 units. Every one of six defects independently forced that result: the signature planner mapped C types incorrectly; rules fired on partial matches and emitted template code; struct definitions in headers were not collected; acceptance was all-or-nothing per project; the C2Rust setup shadowed libc headers; and the harness covered only scalar shapes. Because the v1 corpus was inspected while diagnosing these defects, it is used here only as development data, and the corrected system was evaluated on newly selected libraries under a new protocol.

4.2 Protocol

The v2 protocol fixed the selection rule, configurations, model, acceptance policy, and reporting before any SafeMAP execution on the held-out corpus. A library qualifies if it is a public Git repository with an OSI-approved license, its production code is one `.c` file plus headers that compiles against libc alone, the file has 150–3,500 lines, and it was not used during development. All functions in the file are retained. The protocol and corpus manifest were sealed and the implementation frozen at commit `62cf430` before the corpus was materialized; a materialization script refuses to run on any other commit. The full test suite passed at the freeze.

Four libraries were selected first: parson, sds, tinyexpr, and microtar. After their results were seen, we judged four libraries too few and added six more by the same rule (amendment 1): json-parser, tiny-regex-c, tiny-AES-c, semver.c, argparse, and printf. Nine candidates qualified in a fresh screen; six were chosen for domain diversity before

Table 2 Held-out corpus: ten libraries from two selection rounds.

Library	Round	C LOC	Functions	Units	Ptr. params	Complexity
microtar	Original	376	25	25	39	68
parson	Original	2494	144	138	206	633
sds	Original	1328	45	45	27	292
tinyexpr	Original	1025	48	37	28	306
argparse	Extension	395	13	13	27	101
json-parser	Extension	1186	10	10	15	227
printf	Extension	914	34	20	27	296
semver	Extension	638	33	33	29	127
tiny-AES-c	Extension	572	20	20	33	47
tiny-regex-c	Extension	528	18	15	23	126
Total		9456	390	356	454	2223

any SafeMAP execution on them. The implementation, configuration, and modes were unchanged. We report the two rounds separately and pooled. Table 2 characterizes the corpus.

4.3 Configuration

All LLM configurations, SafeMAP and baseline alike, use Google's `gemini-3.5-flash-lite` at temperature zero through its OpenAI-compatible API, with at most three repairs per unit, 1,000 generated inputs per unit, and seed 20260728. The evaluation used the provider's free tier. A request still refused after the client's retries stops the run rather than being scored as a failure; the run resumes after the quota resets, reusing every finished (project, configuration) result. Because LLM output can vary even at temperature zero, the LLM configurations were run three times on every held-out library with identical inputs; the first run is the primary result. Tool versions: rustc and Clippy 1.99.1, Clang 23.1.3; C2Rust 0.22.1 with LLVM 14 in a pinned container.

4.4 Development data

Development data comprises 40 authored microbenchmarks, six authored modules, ten LLVM test-suite programs pinned at commit `6cdc54e` LLVM Project (2026), and the three v1 libraries. Their results informed the corrections, so they are not evidence of generalization. The component ablations are run on the v1 libraries with the frozen implementation and are reported as development data, which keeps the held-out LLM run within the provider's quota.

4.5 Mutation analysis

To test whether the per-unit check rejects wrong code, we applied single-operator mutations (relational, arithmetic, logical, boolean, and integer-literal changes) to every accepted function of SafeMAP's held-out runs and of its v1-library run, kept the mutants that compile, and re-ran each unit's differential test with 300 inputs.

Table 3 SafeMAP (guided) outcome cascade per library (first run). *Crate*: the unit is in a crate that passed `cargo check`, tests, and Clippy.

Library	Units	Candidate	Generated	Crate	Policy	Accepted
microtar	25	3	2	2	2	2
parson	138	72	4	4	4	3
sds	45	30	4	4	4	2
tinyexpr	37	32	24	0	0	0
argparse	13	0	0	0	0	0
json-parser	10	3	2	2	2	1
printf	20	18	0	0	0	0
semver	33	17	4	4	4	3
tiny-AES-c	20	1	1	1	1	1
tiny-regex-c	15	7	6	6	6	4
Total	356	183	47	23	23	16

5 Results

5.1 RQ1: Accepted units

Table 3 follows each held-out unit through the pipeline. In the first run, of 356 units, the screen marked 183 as candidates, SafeMAP generated Rust for 47, and accepted 16 (4.5% of all units; 8.7% of candidates). The original four libraries contributed 7 accepted units and the extension 9. Accepted units are small leaf functions: a string hash, hexadecimal and UTF-8 helpers, rounding, error strings, version-string checks, regular-expression character classes, and the AES `xtime` step.

Of the 31 generated units not accepted, 24 are from tinyexpr. They compiled, but the crate failed Clippy: two translated constants, `pi` and `e`, keep C's literal digits, which Clippy's default-deny `approx_constant` lint rejects. Because Clippy runs once per crate, this rejected every unit in the crate. We report the outcome as frozen; it is a limitation of crate-level checks within a per-unit policy (Section 7).

5.2 RQ2: Why units are not accepted

Table 4 lists the recorded reasons. Most units never reach translation. 159 functions require manual refactoring, mainly because they read or write mutable global state, perform I/O, or follow allocation protocols that cross function boundaries. 116 use a C type for which the planner finds no safe, testable interface: in parson and sds the central types (`JSON_Value *`, `sds`) are opaque handles or point into the middle of an allocation, and `void *` and `char **` parameters are common. 36 require unsafe operations such as pointer arithmetic beyond a known length or casts between pointer types. Only a small fraction of losses occur after translation: 3 differential-test failures, and 4 units whose C oracle had undefined behavior on most generated inputs. Two of these show a limit of signature-level planning: sds's `sdsll2str` writes up to 21 bytes into a caller buffer, a requirement its signature does not express, so every generated buffer overflowed. The other two overflow signed arithmetic on most inputs.

Table 4 Recorded reasons for functions not accepted by SafeMAP (guided), all ten held-out libraries (first run).

Reason	Functions
Manual refactor (global state, I/O, allocation protocols)	159
No safe, testable interface for a C type	116
Requires unsafe (raw memory, casts)	36
Crate-level Clippy check failed	24
Unsupported construct (union, function pointer, asm)	12
C undefined on most test inputs	4
Differential test failed	3
Callee not translated	3
No synthesis plan	3

Table 5 Accepted units per configuration on the held-out libraries (first run). LLM calls: guided / one-shot. †C2Rust output did not compile.

Library	Units	Guided	One-shot	Rules	C2Rust	LLM calls
microtar	25	2	1	1	0	1 / 1
parson	138	3	4	0	0†	4 / 1
sds	45	2	0	0	0†	4 / 1
tinyexpr	37	0	0	0	0	17 / 1
argparse	13	0	0	0	0	0 / 0
json-parser	10	1	1	0	0	2 / 1
printf	20	0	0	0	0†	0 / 0
semver	33	3	0	0	0	5 / 1
tiny-AES-c	20	1	1	0	0	1 / 1
tiny-regex-c	15	4	2	0	0	6 / 1
Total	356	16	9	1	0	40 / 8

5.3 RQ3: Baselines

Table 5 compares the configurations. SafeMAP accepted 16 units in the first run, the one-shot LLM 9, rules alone 1, and C2Rust none. Over three runs the ranges overlap (Table 8), so we do not claim that SafeMAP outperforms the one-shot baseline. SafeMAP used 40 LLM requests and the one-shot baseline 8 (one per library with planned interfaces). The one-shot baseline benefits from SafeMAP's interface planning: without the planned signatures, its output could not be tested function by function. Raw C2Rust output compiled for seven of ten libraries; for parson, sds, and printf it failed on the pinned toolchain because its `va_list` handling and packed-struct field references are rejected by current rustc. Where it compiled, it contained unsafe code and raw-pointer interfaces throughout, so no unit met the policy.

Development data show the same pattern (Table 6): on the v1 libraries the two accepted 22 and 21 units. Only on the authored microbenchmarks, which the rules were written for, is SafeMAP clearly ahead (72 of 81 units against 35).

Table 6 Development data: accepted units per configuration.

Corpus	Units	Guided	One-shot LLM	Rules only	C2Rust
Microbenchmarks	81	72	35	35	0
Authored modules	25	24	18	18	0
LLVM programs	32	10	8	6	0
v1 libraries	142	22	21	0	0

Table 7 Component ablations on the v1 libraries (development data): accepted / all units.

Configuration	cJSON	inih	libcsv	LLM calls
Full system	15/109	1/10	6/23	23
Facts and plan withheld from prompt	15/109	1/10	6/23	22
Without pointer roles	15/109	0/10	6/23	23
Without safe signatures	2/109	0/10	0/23	126
Without dependency grouping	14/113	1/10	6/23	21
Without idiom plans	3/109	0/10	0/23	3
Without compiler-feedback repair	15/109	1/10	6/23	22

5.4 RQ4: Component ablations

Table 7 reports the ablations on the v1 libraries. The full system accepted 22 of 142 units. Removing safe-signature planning reduced this to 2, while raising LLM requests from 23 to 126, because the model must then invent interfaces that rarely compile with their callers. Removing idiom plans reduced acceptance to 3. Withholding analysis facts and plans from the prompt or removing compiler-feedback repair changed nothing (22), removing dependency grouping lost one unit (21), and removing pointer roles lost one (21). The planning that determines *what* interface each function gets, rather than the guidance given to the LLM, accounts for the accepted translations.

5.5 RQ5: Reliability of acceptance

Mutation analysis. Of 304 compiling mutants of accepted functions (all ten held-out libraries and the v1 libraries), the per-unit check rejected 270 (88.8%). Survivors concentrate in functions whose behavior depends on narrow input ranges: six of the 34 are in sds's hexadecimal-digit converter whose branches are reached only by the few generated inputs that are hexadecimal characters. Some survivors may be equivalent mutants; we did not classify them.

Repeated runs. Table 8 reports accepted units in three runs of the LLM configurations with identical inputs. SafeMAP accepted 16, 15, and 12 units, the one-shot baseline 9, 15, and 14. Eight of the ten libraries gave the same SafeMAP result in every run. Most of the variation comes from semver: in one run of each configuration, one generated function failed a crate-level check and so rejected all four planned units of that crate, the same mechanism as tinyexpr. The accepted sets are therefore stable at the function level but sensitive to crate-level failures.

Table 8 Accepted units across repeated LLM runs (identical inputs and configuration).

Run	Libraries	Guided	One-shot LLM
Run 1	10	16	9
Run 2	10	15	15
Run 3	10	12	14

6 Discussion

Coverage is the limiting factor. The per-function acceptance check is feasible on production C and rejects most wrong code, but few functions reach it. Table 4 is the main finding: the obstacles are global state, opaque and aliasing pointer types, and raw memory manipulation. These are API design questions that a function-local translator cannot answer alone; a migration would need to change the library's interface, for example by turning global state into an explicit context or an `sds` string into an owned type, before SafeMAP's checks apply.

Interfaces matter more than prompts. Given the planned interfaces, a single LLM request performs within the run-to-run range of SafeMAP's guided loop; without them, the guided loop collapses (2 of 142 units). Withholding analysis facts from the prompt changed nothing on the v1 libraries. This suggests that LLM translators benefit most from being told the target interface, which is consistent with CRUST-Bench's use of hand-written interfaces Khatry et al. (2025).

Refusal as output. Every refused unit carries a reason and the plan that produced it. For a migration team this identifies which functions can be migrated mechanically and which need design work.

7 Threats to validity

Sample. Ten small libraries with one production file each do not represent large applications or multi-file projects. The extension was decided after the first four results were seen; it used the same rule and frozen code, and we report both rounds separately.

Model and variance. One low-cost model was used, and three runs are too few to estimate run-to-run variance precisely; the guided and one-shot ranges overlap, so their difference is not established. A stronger model may translate more of the planned units, although it cannot change the units refused before translation, which are most of the losses.

Oracle. Differential testing on generated inputs is evidence, not proof. Mutation analysis shows that 11.2% of compiling mutants survive, mainly where inputs rarely reach a branch. Excluding inputs with undefined C behavior removes inputs the sanitizers detect, not all undefined behavior.

Crate-level checks. Clippy runs on the whole crate, so one lint error rejects every unit in it (tinyexpr). This makes the reported numbers conservative.

No eligibility labels. We did not collect independent labels of which functions could be migrated safely, so we make no claim about the precision or recall of the screen; refusals are reported as recorded.

Toolchain. C2Rust 0.22.1's output does not compile with the pinned nightly on three libraries; another toolchain might compile it, but it would still use unsafe code and fail the policy.

Development data. The development corpora and the v1 libraries shaped the implementation; results on them are not evidence of generalization, and the ablations are run on them.

8 Conclusion

SafeMAP translates C functions to safe Rust only when it finds a safe interface and accepts them only when they match the original C under sanitizers. On ten held-out libraries it accepted 12–16 of 356 units across three runs, against 9–15 for a one-shot LLM given the same planned interfaces, 1 for its rules alone, and 0 for raw C2Rust. The interface planning, not the LLM guidance, is what makes these translations possible to produce and check. Most units are refused before translation because of global state, opaque or aliasing pointer types, or unsafe memory operations. Per-function, oracle-checked migration is therefore feasible but narrow: its reach is set by how much of a C library's interface already has a safe Rust counterpart. Future work should plan such interface changes explicitly, generate inputs that target narrow branches, and run lint checks per unit.

Statements and Declarations

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Competing interests

The authors declare no competing financial or non-financial interests relevant to the content of this article.

Data availability

The source code, evaluation protocol, corpus manifests, configuration, frozen implementation record, raw results, and scripts that generate every table are available in the SafeMAP repository (https://github.com/jscyril/SafeMAP); the implementation evaluated here is commit `62cf430`. The result rows of every run are in `reports/publication/v2`, and `scripts/generate_v2_manuscript_tables.py` regenerates every table from them. The held-out libraries are public repositories pinned to the revisions recorded in the manifests.

Ethics approval

Not applicable. The study involved no human participants.

Use of generative AI

Generative AI tools were used in this work. ChatGPT assisted with generating the synthetic development benchmarks and with language refinement. Claude assisted with implementing and testing the software, writing the evaluation scripts, and drafting and revising the manuscript. The evaluated pipeline itself calls Gemini as described in the paper. The authors reviewed and validated all AI-assisted code, data, and text, and remain responsible for the accuracy, originality, and integrity of the manuscript.

Author contributions

Jacob Sebastian Cyril: conceptualization, methodology, software, investigation, formal analysis, visualization, and writing—original draft. Benita Jose Chalissery: research guidance and manuscript review. NewBegin M: supervision and manuscript review. All authors approved the final manuscript.

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